MIL-E-1006D

5 August 1968
SUPERSEDING
MIL-E-1006C
8 February 1965

MILITARY SPECIFICATION

EMBALMING FLUID, ARTERIAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and grade of arterial embalming fluid (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-40030 - Drums, Plastic, Molded Polyethylene.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48", Pallet, Skids, Runners, or Pallet-Type Base.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9930

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* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) DESIGNATION:

D 1193-66 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

AGENT T. C. GEORGE'S TARIFF NO. 15

Interstate Commerce Commission Regulations

(Application for copies should be addressed to T. C. George, Agent, 63 Vesey Street, New York, New York 10007).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials.- The manufacturer is given wide latitude in the selection of materials for use in formulating the arterial embalming fluid provided the product conforms to the requirements of this specification.

* 3.2 Composition.- The following ingredients shall be present in the finished formulation in the amounts indicated in table I when tested as specified in 4.3.4. These amounts are expressed in percent by weight, except for formaldehyde index number which is expressed in grams of formaldehyde per 100 milliliters of solution.

TABLE I.- Composition

Ingredients	Minimum	Maximum
Formaldehyde index number	25.0	28.0
Blood dissolving compound (sodium tetroborate, dodecahydrate or equivalent)	0.5	6.5
Glycerine or sorbitol	8.0	10.0
Alcohols 1/	8.0	9.0
Wetting agent (Triton X-100 or equivalent)	0.5	---

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1/ The only alcohols permitted in this formulation are methanol, ethanol, isopropanol, and ethyl alcohol denatured with methanol.

3.2.1 Prohibited ingredients.- The arterial embalming fluid shall contain no antimony, arsenic, lead, mercury, copper, zinc, silver, chloral, cyanide, nor alkaloids, in excess of 5 parts per million, when tested as specified in 4.3.2.

3.3 Color.- The arterial embalming fluid shall contain a dye as required to give the fluid a distinct red or pink color when examined visually (see 4.3.1.1).

3.4 Workmanship.- The arterial embalming fluid shall be clear, homogeneous, and free from sediment.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter. For purposes of sampling, and inspection lot for examinations and tests shall consist of all the arterial embalming fluid prepared from the same materials, under essentially the same conditions, and submitted for inspection at one time.

4.2.1 Inspection of materials and components.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.-

4.3.1 Examination of the end item.- Each lot of end items shall be examined for the defects in the applicable subparagraphs at the inspection levels and the acceptable quality levels (AQLs) set forth in 4.3.3. The lot size, for purposes of determining the sample size in accordance with

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MIL-STD-105, shall be expressed in units of filled containers for the examinations in 4.3.1.1 and 4.3.1.2, in units of shipping containers for the examination in 4.3.2.1, and in units of pallets for the examinations in 4.3.2.2.

* 4.3.1.1 Examination of the packaged end item.- The sample unit for this examination shall be one container of arterial embalming fluid.

Examine	Defect
Package	Not as specified. Closure not as specified. Any leakage.
Workmanship 1/	Liquid not clear; sediment. Not homogeneous.
Color 1/	Not a red or pink color in reflected light.

1/ The sample units shall be opened and poured out into clear glass containers to determine compliance.

4.3.1.2 Examination of the unit container for net contents.- The sample unit for this examination shall be one container of embalming fluid. The lot shall be unacceptable if the average volume content per unit container is less than the specified quantity.

4.3.2 Examination of preparation for delivery.-

* 4.3.2.1 Examination for packing and marking.- An examination shall be made to determine that packing and marking comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery except that it shall not be palletized and need not be sealed. Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure. The lot size shall be number of shipping containers in the end item inspection lot.

Examine	Defect
Packing	Not in accordance with contract requirements. Shipping container not as specified; closures not accomplished by specified or required methods or materials. Bulged or distorted container.
Marking	Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

* 4.3.2.2 Examination for palletization.- An examination shall be made to determine that palletization complies with the requirements of Section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.

4.3.3 Inspection levels and acceptable quality levels (AQL's) for examinations.- The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL
4.3.1.1	I	2.5
4.3.1.2	S-2	N/A
4.3.2.1	S-2	2.5
4.3.2.2	S-1	6.5

4.3.4 Testing of the end item.- The methods of testing listed in table II shall be followed for each lot. The lot size shall be expressed in gallons of embalming fluid. Each lot shall consist of a single composition of product manufactured under essentially the same conditions, and at essentially the same time. The sample unit for testing shall be (1) pint composite, obtained by randomly selecting and thoroughly mixing one-half pint samples in the frequency indicated below. The composite shall be placed in a clean, dry, sealed pint container. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, weighing and storage. The lot shall be unacceptable if the composite fails to meet any of the test requirements specified.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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INSTRUCTIONS FOR TESTING OF THE END ITEM

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations Per Unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Comp Samp	Samp Unit		Pass or Fail	Numerically to Nearest		
Formaldehyde index number	Table I	4.4.1	X		2		Average of 2 determinations, to nearest 0.01 number		
Blood dissolving compound (sodium tetraborate, dodecahydrate or equivalent)	Table I	1/							
Glycerine or sorbitol	Table I	1/							
Alcohols	Table I	1/							
Wetting agent (Triton X-100 or equivalent)	Table I	1/							
Prohibited ingredients	3.2.1	1/							

1/ Certificate of compliance shall be furnished the contracting officer for the indicated composition requirements.

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8 February 1965

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4.4 Test procedures.-

4.4.1 Formaldehyde index number.-

* 4.4.1.1 Method of analysis.- This method of analysis is based on the reaction of formaldehyde with sodium sulfite to give free caustic soda and the sulfitealdehyde addition product. The reaction is specific, rapid and quantitative.

$$CH_2O + Na_2SO_3 + H_2O \longrightarrow NaOH + CH_2 (NaSO_3) OH$$

NOTE: In the handling, weighing, and storing of embalming fluids, take the necessary precautions to keep the loss of volatile ingredients to a minimum.

* 4.4.1.2 Reagents.-

a. Sodium sulfite solution, 1 molar concentration, prepared by dissolving 252 grams of hydrated sodium sulfite ($Na_2SO_3 \cdot 7H_2O$) or 126 grams of andydrous sodium sulfite (Na_2SO_3) in distilled water and make to one liter.

b. Distilled water, conforming to the requirements of ASTM D1193-66, Reagent Water.

c. Sodium hydroxide solution, 0.5 normal, standardized to thymolphthalein end point (disappearance of blue color; pH 9.0 to 9.3).

d. Sulfuric acid solution, 0.5 normal, standardized to thymolphthalein end point.

e. Thymolphthalein indicator solution, a 0.1 percent solution in 90-95 percent ethyl alcohol.

4.4.1.3 Equipment.-

a. pH meter.

* 4.4.1.4 Procedure.- Fifty milliliters (ml) of molar sodium sulfite solution and 5 drops of thymolphthalein indicator solution are placed in a 150 ml beaker and carefully neutralized by titration with standard sulfurci acid solution to an end point (the disappearance of the blue color). When neutralization is completed, take pH of solution with carefully standardized pH meter. Accurately weigh the beaker and neutralized solution to the nearest milligram. Pipette 3 ml of the arterial embalming fluid into the neutralized sulfite solution and immediately and rapidly weigh beaker and contents to determine the weight (sample weight, S) of embalming fluid added. Stir the solution for 1 minute, the immediately titrate to the previous pH of the solution, using the standard 0.5N sulfuric acid solution. Record the number of milliliters (A) of sulfuric acid required.

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Prepare a blank consisting of 3 ml of the embalming fluid and 5 drops of thymolphthalein indicator solution in 50 ml of distilled water and immediately titrate the solution with standard 0.5N sodium hydroxide solution to the previously determined pH value. Record the number of ml (B) of standard 0.5N sodium hydroxide solution required to obtain this pH.

Calculate the percent formaldehyde in the test embalming fluid and the formaldehyde index number as follows:

$$\text{Percent formaldehyde} = \frac{|(A \times N_A)| + (B \times N_B) \times 3.003}{S}$$

Where:

A = ml of standard sulfuric acid solution required in titration of test sample.

B = ml of standard sodium hydroxide solution required in titration of blank.

N_A = normality of standard sulfuric acid solution.

N_B = normality of standard sodium hydroxide solution.

S = sample weight (grams) of test embalming fluid.

Formaldehyde index number = specific gravity of embalming fluid determined at 20°C. multiplied by percent formaldehyde (as determined).

NOTE: The specific gravity at 20°C. of the embalming fluid shall be determined accurately to the second decimal place.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.- Fifteen gallons of embalming fluid shall be unit packaged in a polyethylene drum conforming to style B, size 2B of MIL-D-40030. Each polyethylene drum shall be closed in accordance with the appendix of MIL-D-40030.

5.1.2 Level C.- The embalming fluid shall be packaged to afford adequate protection against physical deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B or C as specified (see 6.2).

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* 5.2.1 Level A.- Each 15 gallon polyethylene drum of embalming fluid, packaged as specified in 5.1, shall be overpacked in a steel drum in accordance with ICC-6D or ICC-37M.

* 5.2.2 Level B.- Each 15 gallon polyethylene drum of embalming fluid, packaged as specified in 5.1, shall be overpacked in a steel drum conforming to ICC-6D or ICC-37M.

5.2.3 Level C.- The embalming fluid, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

* 5.3 Palletization.- Unless otherwise specified (see 6.2), embalming fluid, packed as specified in 5.2., shall be palletized in accordance with load type and XIII of MIL-STD-147. Storage aids 5 and 7 shall apply, as applicable. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means F, K, and L, as applicable.

5.4 Marking.- In addition to any special marking required by the contract or order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

* 5.4.1 Special marking.- The following special marking shall be shown on the label of each shipping container:

"EMBALMING FLUID, ARTERIAL
FLAMMABLE
POISON
FORMALDEHYDE INDEX NO.
CONTRACT NO.
NAME OF MANUFACTURER
MANUFACTURER'S DIRECTIONS FOR USE

Do not use in presence of open flame.

Keep container tightly sealed when not in use to prevent vaporization and loss of strength. Dilute as required with water."

6. NOTES

6.1 Intended use.- The arterial embalming fluid covered by this specification is intended for mortuary use in the preparation of human remains for burial.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) When palletization is not required (see 5.3).

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6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - MS
Air Force - 34

Review activities:

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Navy - MS
Air Force - 34

User activities:

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Air Force - 26

Preparing activity:

Army - GL

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